

SYLARNA, Sweden

WMO: 022090

Lat: 63.0438N

Lon: 12.2748E

Elev: 1030

StdP: 89.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.3	-18.7	-28.1	0.3	-19.9	-23.8	0.5	-16.5	23.2	-6.6	21.2	-6.3	3.5	160	0.948

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	18.9	12.2	17.1	11.6	15.4	10.7	13.6	17.5	12.5	15.9	11.4	14.5	4.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	9.8	15.1	10.6	9.0	14.0	9.6	8.4	13.2	41.4	17.5	38.3	15.9	35.4	14.6	17.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.4	14.5	DB	-21.9	21.1	5.7	2.0	-26.0	22.5	-29.3	23.7	-32.6	24.8	-36.7	26.3	(4)
(5)			WB	-21.7	14.9	5.4	1.7	-25.6	16.1	-28.7	17.1	-31.7	18.1	-35.6	19.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.3	-6.8	-7.8	-6.1	-1.6	2.4	6.6	9.7	9.1	5.5	0.2	-3.4	-5.2
	DBStd	7.45	4.85	5.87	4.43	3.92	4.20	3.94	3.66	3.53	3.55	3.93	4.82	4.80
	HDD10.0	3651	522	497	500	349	238	116	50	61	141	305	402	471
	HDD18.3	6597	780	731	758	599	493	352	267	287	385	563	652	729
	CDD10.0	96	0	0	0	0	3	14	42	32	6	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.6	8.2	7.9	7.5	6.4	5.7	5.4	4.9	5.2	6.2	6.9	7.3	7.9
Precipitation	PrecAvg	808	67	55	55	37	49	82	102	94	83	69	58	57
	PrecMax	1043	149	136	117	101	85	123	181	151	187	116	157	128
	PrecMin	561	3	2	20	10	26	32	34	35	27	24	2	12
	PrecStd	101	40	29	29	22	15	25	38	35	44	22	43	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	5.6	4.9	8.6	17.2	19.3	22.4	20.4	15.9	11.6	8.2	6.0
		MCWB	1.3	1.9	1.1	4.8	10.7	12.1	13.9	12.9	10.6	8.5	4.0	2.5
	2%	DB	1.8	2.4	2.7	6.9	13.4	16.9	19.7	18.1	13.8	8.1	6.0	3.9
		MCWB	-0.4	-0.3	0.5	3.7	8.2	10.5	12.6	12.1	9.6	5.6	3.5	1.5
	5%	DB	0.3	0.3	1.1	5.3	10.7	14.6	17.6	16.3	12.2	6.7	4.0	2.4
		MCWB	-1.3	-1.2	-0.5	2.5	6.6	9.5	11.9	11.5	9.0	4.4	2.4	0.5
	10%	DB	-1.0	-1.3	-0.4	3.8	8.6	12.8	15.7	14.5	10.7	5.4	2.4	0.7
		MCWB	-2.2	-2.6	-1.7	1.4	5.2	8.7	11.0	10.7	8.1	3.4	0.8	-0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	2.6	2.2	5.9	11.3	13.6	15.1	14.7	11.8	9.3	5.5	3.7
		MCDB	3.3	4.9	3.6	7.7	15.7	16.6	20.1	18.1	14.1	11.1	7.2	5.2
	2%	WB	0.3	0.2	0.9	4.0	8.6	11.3	13.9	13.2	10.6	6.1	3.9	2.0
		MCDB	1.6	1.5	2.2	6.4	12.6	15.0	18.5	16.5	12.7	7.6	5.4	3.5
	5%	WB	-0.9	-1.2	-0.5	2.7	6.9	10.1	12.9	12.2	9.4	4.9	2.4	0.5
		MCDB	0.2	0.0	0.9	4.8	10.1	14.0	16.7	15.4	11.6	6.5	3.8	2.0
	10%	WB	-2.1	-2.5	-1.6	1.7	5.5	8.9	11.7	11.1	8.2	3.7	0.9	-0.8
		MCDB	-1.2	-1.4	-0.4	3.5	8.4	12.5	15.1	14.1	10.3	5.2	2.3	0.4
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.0	4.9	4.6	5.5	6.2	6.2	5.4	4.9	4.1	4.5	4.8
		MCDBR	5.5	6.2	5.7	5.4	8.2	9.3	9.3	7.8	7.0	5.7	5.2	6.3
		MCWBR	4.7	4.7	4.7	3.9	5.3	5.6	5.6	4.7	4.6	4.6	4.4	5.2
	5% WB	MCDBR	5.3	5.8	5.5	5.1	7.8	8.3	8.2	7.0	6.0	5.5	5.3	6.1
		MCWBR	4.8	4.7	4.7	3.9	5.5	5.6	5.5	4.7	4.5	4.7	4.8	5.2
Clear-Sky Solar Irradiance	taub	0.181	0.239	0.266	0.292	0.304	0.304	0.314	0.304	0.281	0.261	0.190	0.162	
	taud	2.036	2.257	2.269	2.298	2.453	2.538	2.521	2.539	2.590	2.479	2.090	1.955	
	Ebn at Noon	502	735	848	896	915	922	901	883	844	714	475	292	
	Edh at Noon	29	56	83	101	96	90	89	80	62	46	27	15	
All-Sky Solar Radiation	RadAvg	0.19	0.76	2.15	3.99	4.68	4.93	4.67	3.57	2.18	0.95	0.27	0.10	
	RadStd	0.02	0.11	0.22	0.31	0.33	0.44	0.53	0.35	0.22	0.12	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-142	-178	N/A	N/A

Nomenclature: See separate page